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COMPARING THE PERFORMANCE OF HIMAWARI-8 SATELLITE AND GEO-KOMPSAT 2A SATELLITE IN ACTIVE WILDFIRE DETECTION OVER EASTERN AUSTRALIA

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ABSTRACT

One of the most severe bushfire in history took place in eastern Australia between 2019 and 2020, evoking worldwide attention. Utilizing satellite data for active fire detection is essential for safeguarding the ecological environment and maintaining socioeconomic stability. This article compares and analyzes the performance of two commonly used geostationary satellite data, Himawari-8 and Geo-KOMPSAT 2A (GK2A), in active fire detection over eastern Australia. The active fire detection based on the random forest model has high accuracy and robustness to features. The results show that the precision of GK2A satellite data in active fire detection is 0.83, with the omission of 0.26 and the F1-score of 0.78. The precision of Himawari-8 is 0.68, with the omission of 0.35 and the F1-score of 0.66, indicating that GK2A satellite is more feasible than Himawari-8 satellite in active fire detection over this region.

***Index Terms*—** active fire detection, random forest model, Himawari-8, Geo-KOMPSAT 2A (GK2A), precision, omission, F1-score

1. INTRODUCTION

As the world's industrialization continues to accelerate, the number of global wildfires has continued to increase since the 1950s [1]. Wildfires can cause serious ecological damage and economic losses, threaten human life and property, destroy land vegetation and topsoil, and affect many countries and regions around the world [2-4]. Many countries, such as China, the United States, and Australia, are plagued by wildfires every year. It is for this reason that many countries attach great importance to fire detection. Many different satellite data are used for fire spot detection, such as Landsat-8 Landsat, Himawari-8 and HJ-IRS [5-9]. However, in general, only one satellite series is used for active fire detection, so it is important to compare the feasibility of different satellite series for active fire detection.

Nowadays, many active fire detection studies focus on Landsat and Sentinel satellites, such as Y He using Landsat 8 data for active fire detection, Gargiulo's CNN-based super-resolution technology for active fire detection on Sentinel-2 data, and White comparing Sentinel 2 and Fire extent and severity mapping between Landsat 8 satellite sensors, Konkathi used Google Earth Engine to compare the post-fire burn severity index of Landsat-8 and Sentinel-2 images to each other [5,10-11]. However, these two series of satellites are difficult to solve the cloud coverage problem in active fire detection [13].

The Himawari 8 satellite operated by the Japan Meteorological Agency (JMA) and the Korean Geostationary Korea Multi-Purpose Satellite 2A (Geostationary Korea Multi-Purpose Satellite 2A) developed and operated by South Korea can effectively solve the problems of cloud occlusion and cloud coverage [14,15]. Therefore, based on the random forest model, the feasibility analysis and comparison of the Sunflower 8 satellite/Geo-KOMPSAT 2A satellite in active fire detection provides a new reference value for global active fire detection.

2. MATERIALS

2.1 Study area

Between 2019 and 2020, Australia experienced a massive forest fire, which became the focus of global attention at that time. The fires spread rapidly, covering millions of hectares of land, destroying thousands of buildings and killing dozens of people. The fires have burned through a record area of forest in southeastern Australia since European settlement, surpassing even the massive Black Friday fires of 1939.

The study area for this study is located in eastern Australia and has a diverse topography with numerous hills and mountains. The climate in the eastern region is generally mild, but has considerable seasonal variation. The overall pre-precipitation in the study area is low, the climate is relatively dry in spring, and wildfires are prone to occur. Forests are abundant, including tropical rainforests, eucalyptus forests, and coniferous forests [16]. For these reasons, eastern Australia is a high-risk area for fires. Fires occur frequently,especially during the dry season, and forests and grasslands are often threatened by fires, which have a huge impact on local ecosystems and wildlife.

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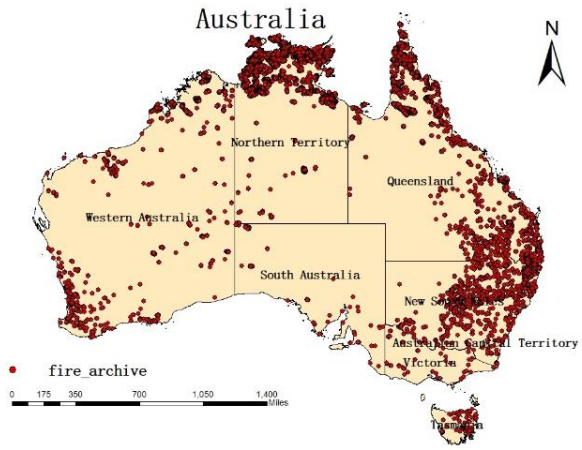

Fig 1. Distribution of fire spots in Australia from August to September 2019.

2.2 Himawari-8 netCDF4 Data

Himawari 8 is a meteorological satellite operated by the Japan Meteorological Agency (JMA). It is a geostationary meteorological satellite. It is the first satellite in the Sunrise series and was successfully launched by the H-IIA launch vehicle on October 7, 2014. Table 1 shows the specifications of the AHI (Advanced Himavari Imager) imager it carries, with a spatial resolution ranging from 500 m to 2 km, covering areas such as East Asia and Australia [14]. Additionally, the data provided by Himawari-8 is in NetCDF (Network Common Data Format) format. The data used in this study are brightness temperatures in the infrared band (band 7-16).

Table 1. Himawari-8 AHI Imager specifications.

Type	Band	Wavelength (μm)	Spatial Resolution (km)
Visible	1	0.47	1
	2	0.57	
	3	0.61	0.5
Near-infrared	4	0.86	1
	5	1.6	2
	6	2.3	
Infrared	7	3.9	
	8	6.2	
	9	6.9	
	10	7.3	
	11	8.6	
	12	9.6	
	13	10.4	
	14	11.2	
	15	12.4	
	16	13.3	

2.3 GEO-KOMPSAT-2A netCDF4 Data

Geo-KOMPSAT 2A (Korea Geostationary Multipurpose Satellite 2A) is a geostationary multipurpose satellite provide a wide range of Earth observation and meteorological monitoring services. On December 4, 2018, during a launch mission in French Guiana, an Arian 5 rocket successfully launched Geo-KOMPSAT 2A. Table 2 shows the specifications of the AMI imager it carries. The GK2A AMI can have a resolution of 0.5~2 kilometers every 2 minutes around the Korean Peninsula, with a total area of 10 minutes [15]. Similarly, the data provided by GK2A is also in NetCDF format. This study uses the brightness temperature of the shortwave infrared channel (channel 7), water vapor channel (channel 8-10) and infrared channel (channel 11-16).

Table 2. GEO-KOMPSAT-2A AMI Imager specifications.

Type	Band	Wavelength (μm)	Spatial Resolution (km)
Visible	1	0.47	1
	2	0.51	
	3	0.64	0.5
	4	0.86	1
Near-infrared	5	1.4	2
	6	1.6	
Shortwave -infrared	7	3.8	
Water vapor	8	6.2	
	9	6.9	
	10	7.3	
Infrared	11	8.6	
	12	9.6	
	13	10.4	
	14	11.2	
	15	12.4	
	16	13.3	

2.4 Fire point label data

In this study, MCD14ML product was used as the fire point dataset for model training and testing, which is a thermal anomaly/fire location product produced by a Moderate Resolution Imaging Spectroradiometer (MODIS) and has been updated to the sixth generation. The fire point product is processed by NASA's Scientific Computing Facility (SCF) at the University of Maryland (UMD) and distributed by the Fire Information Module (FIRMS) in the resource management system. This product was synthesized using a L2 thermal anomaly product (MOD14/MYD14). Thermal anomalies/active wildfires are marked in the center of 1 km pixels, and the identification principle of the wildfire fire points is based on the MOD14/MYD14 fire and thermal anomaly algorithm, and one or more fire times may be included in the marked pixels. This product is the most basic point of fire product for identifying wildfire fires and other thermal anomalies such as volcanoes.

3. METHOD

3.1 Data Preprocessing

The data used in this study are the 7-16 channels of Sunflower 8 and Geo-KOMPSAT 2A. Because the original data has 16 channels, the original data of two different satellites were extracted from the 7-16 channel data and passed the official calibration The brightness temperature of each channel can

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be obtained by looking up the table. According to the conversion table given on the official website, the digital count of each channel is converted into open temperature storage. Since the original data used is full-size image data, it needs to be cropped to within the study area of this article. In order to make edge pixel values more accurate during cropping, the following strategy can be adopted: for sites close to pixel vertices, take the average of the last four pixels or take the average of the pixel values of the last nine pixels [17].

3.2 Construction of Fire Point Dataset

This study selected remote sensing image data and existing wildfire products corresponding to the fire outbreak time in the study area from 2020 to 2021 to construct a fire point data set. The wildfire data in the dataset comes from the MCD14ML product. Integrate the format data, time information, and various ignition tags obtained by preprocessing of the Sunflower-8/GK2A data set package. The specific production process is as follows:

1. Extract MODIS ignition point. Extract ignition point data for the same time period, extract all ignition points of the day from MCD14ML, and combine non-vegetation ("type field value is 1"), non-daytime ("day and night field value is N"), low confidence ("confidence field value Less than 70") ignition point is regarded as "MODIS ignition point".
2. Through search, the Sunflower-8/GK2A time period with "MODIS" was determined, and the corresponding MODIS data was downloaded according to the corresponding time period, and cropped to the research area of this study, while matching various fire points and other information.
3. Randomly select 10x vegetation and low-confidence ignition points as “non-MODIS ignition points”

Then, all samples were divided into test set and training set at a ratio of 1:4. Brightness temperature information extracted for each sample pixel.

3.3 Random Forest

The random forest model was proposed by Brayman in 2001 [18] and has been applied to remote sensing wildfire detection [19]. Random forest belongs to the bagging (abbreviation of Bootstrap AGgregation) method in ensemble learning. When performing a classification task, new input samples enter, and each decision tree in the forest is judged and classified separately, and each decision tree will get its own classification. In this way, which classification result of the decision tree has the most classifications, then the random forest will use this result as the final result. Compared with other machine learning models, the random forest model has good overfitting robustness.

First, the constructed fire point data set is divided into a test set and a training set at a ratio of 1:4; then the data set is input into the random forest model for training. Since the number of positive samples in the data set is relatively small, an attention mechanism is introduced during the training process to give higher weight to the fire samples. Pixels classified as fire by the random forest model are labeled as potential fire pixels [20].

In addition, according to Planck's theorem and Stilpan-Boltzmann's theorem, combined with the temperature characteristics of vegetation burning, the peak energy release during vegetation burning is about 3.8 μm, corresponding to the 7th band of Sunflower-8/GK2A . Conversely, wildfires emit lower energy in the thermal infrared as they burn, while the difference between mid- and thermal infrared can eliminate wildfires and other high-temperature features (e.g., man-made features) that emit higher in both the mid- and thermal infrared energy of). Therefore, pixels that meet condition 1 will also be marked as potential triggering pixels.

$$T_{b7} > 300K \ and \ T_{b7} - T_{b14} > 10K \qquad (1)$$

T_{b7} represents the brightness temperature of Himawari-8/GK2A in the 7-band band, and T_{b7}-T_{b14} represents the difference between the brightness temperature of the 7-band and the 14-band of the thermal infrared channel (TIR) in Himawari-8/GK2A data. The two thresholds of 300 and 10 are determined according to the burning intensity of the wildfire in the study area and the energy released by the burning area, which belong to the empirical thresholds.

3.4 Accuracy evaluation methods and indicators

According to the fire point information generated by the model, the generated fire point should be compared with the fire point label of the original data to evaluate the performance of the model. According to the experimental results, the confusion matrix is constructed [21]. The confusion matrix has four important parameters TP (true examples), FN (false negative examples), FP (false positive examples), TN (true negative examples).

In the image, if the coordinates of the predicted fire point and the coordinates of the fire point label are located in the same cell, the fire point is a real example. If the location of the predicted fire point is not labeled, the predicted fire point is a false positive example. If The model did not successfully detect the fire point in the label, the location is a false negative example and the remaining pixels are true negative examples.

On the basis of the confusion matrix, the Precision, Omission (equal to 1-Recall) and F1-score are calculated, and the calculation methods are shown in Equations 2, 3, and 4 respectively.

$$Precision = \frac{TP}{TP + FP} \qquad (2)$$

$$Omission = \frac{FN}{TP + FN} \qquad (3)$$

$$F1 = \frac{2 * Precision * (1 - Omission)}{1 + Precision - Omission} \qquad (4)$$

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4. RESULTS

Table 3 shows some accuracy indicators of active fire point detection using Himawari-8/GK2A satellite data under the random forest model method. The results show that under the random forest model method, the accuracy of GK2A satellite data in active fire detection is 0.83, which is significantly higher than the accuracy of Himawari-8 of 0.68, the missed detection rate of 0.26 is lower than the missed detection rate of Himawari-8 of 0.35, and the F1-score is 0. 78 is higher than Himawari-8's F1-score0.66, which shows that using the random forest model method, GK2A satellite data is more feasible than Himawari-8 data in active fire detection.

Table 3. Accuracy evaluation of the data obtained from Himawari-8/GK2A.

Data to Be Evaluated	Precision	Omission	F1-score
Data obtained from Himawari-8 data	0.68	0.35	0.66
Data obtained from GK2A data	0.83	0.26	0.78

5. CONCLUSION

In this study, the feasibility of Himawari-8 satellite data and GK2A satellite data in active fire detection was compared, and the results showed that GK2A satellite data was more accurate than Himawari-8 data in active fire detection, lower missed detection rate, higher F1-score, and more feasible under the method of random forest model. Next work will focus on further evaluating the feasibility of Himawari-8 satellite data and GK2A satellite data for active fire detection using other methods.

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